CSC186 – Object Oriented Programming
Academic Session October 2025 – February 2026
Lab Assignment 1 - Basic Programming

Course Outcomes (CO)	LO1	LO2	LO3
CO1			
CO2	√	√	√
CO3			

Write a program using Java Programming Language
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QUESTION 1

Keretapi Tanah Melayu Berhad (KTMB) operates an inter-city rail service known as Electric Train Service (ETS) to several destination of the West Coast Line between Gemas and Padang Besar in Malaysia. As a new programmer positioned in KTMB Penang Branch, you are assigned to develop an application program to calculate the total price of ETS ticket for each passenger that commute using this service from Butterworth to Kuala Lumpur. The ticket price for each package is as follows:

Type of Package	Ticket Category	Ticket Price (RM)
ETS Gold (G)	Adult	59
	Child	34
ETS Platinum (P)	Adult	79
	Child	44

Write a complete Java application program to perform each of the following:

a) Prompt the user to input the following information:
- Passenger ID (using a string input)
- Type of package (using the character code, G or P)
- Quantity of adult passengers
- Quantity of child passengers

The program should display appropriate message if the user has entered an invalid package code.

b) If the passenger wanted to add on a meal combo, prompt the user to input the following information:
- Add-on meal combo (yes or no)
- Quantity of meal combo

RM 7 is added up for each meal combo requested. Otherwise, the ticket price remains the same. Then, calculate the total price for each passenger ID, meal combo price and the net price for each passenger ID including the meal combo (if requested).

c) Display the passenger information as shown in the following example:

```
ETS TICKET INFORMATION - Butterworth to Kuala Lumpur


PASSENGER ID                A123456

Number of Adults            2
Number of Children          3

Total Ticket Price          RM 220.00
Meal Combo Price            RM 35.00
Net Ticket Price            RM 255.00
```

d) The program should repeat the steps in (a) to (c) until the user enters 'n' to stop.

QUESTION 2

The Entrepreneur Club of UiTM Cawangan Pulau Pinang will be organizing the entrepreneurship week after the final examination. The organizer offers canopies to be rented for the students, staffs and outsiders who are interested to join the events. As a programmer you have been appointed to write a program to calculate the total charges of rental that should be paid by the participants.

a) By referring to the following table of canopy charges, write a function named `canopyCharges (...)` to calculate the canopy charges which receives the parameter of participant code, total of canopies and total days. Return the calculated charges to the main program.

Participant Type	Participant Code	Charges/Day
Students	T	RM90.00
Staffs	S	RM150.00
Outsiders	O	RM200.00

b) Write a function named `additionaiCharges(...)` which receives the item code from the main program. The participants need to pay an additional of RM 100 if they decided to sell the wet (W) items. Otherwise, no charges will be incurred.

c) Write A Java application program which performs the following tasks.

i) Read the participant code, total of canopies, total days of rental and item code.
ii) Call the function named `canopyCharges(...)` and send the participant code, total of canopies and total days of rental to calculate the total charges.
iii) Call the function named `additionaiCharges(...)` and send the item code to calculate the total of additional charges (if incurred).
iv) Calculate and display the total charges of canopies and additional charges.
v) The program will be called repeatedly until user requests to stop.

ANSWERS

Questions 1

```
import java.util.Scanner;

public class KTMB {
  public static void main(String[] args) {
    Scanner keyboard = new Scanner(System.in);
    boolean isRepeat;
    do {
      String passengerID;
      char packageCode;
      int adultQuantity = 0;
      int childQuantity = 0;

      double ticketPrice;
      final double ADULT_GOLD_PRICE = 59.00;
      final double CHILD_GOLD_PRICE = 34.00;
      final double ADULT_PLATINUM_PRICE = 79.00;
      final double CHILD_PLATINUM_PRICE = 44.00;
      double total;

      boolean addOn;
      final double MEAL_COMBO_PRICE = 7.00;
      int mealComboQuantity = 0;
      double mealComboCharge = 0.00;

      // Get Pessenger ID
      System.out.print("\nPlease enter Passenger ID: ");
      passengerID = keyboard.nextLine();

      // Table of Prices
      System.out.println("\n| Type of Package  | Ticket Category | Ticket
Price (RM) |");
      System.out.println("| ---------------- | --------------- | -------------
---- |");
      System.out.printf("| ETS Gold (G)     | Adult           | %.2f
|\n", ADULT_GOLD_PRICE);
      System.out.printf("| ETS Gold (G)     | Child           | %.2f
|\n", CHILD_GOLD_PRICE);
      System.out.printf("| ETS Platinum (P) | Adult           | %.2f
|\n", ADULT_PLATINUM_PRICE);
      System.out.printf("| ETS Platinum (P) | Child           | %.2f
|\n", CHILD_PLATINUM_PRICE);

      // Get Package Code
      System.out.print("Please enter package code (G/P): ");
      packageCode = Character.toUpperCase(keyboard.next().charAt(0));

      // For invalid Package Code
      while (!(packageCode == 'G' || packageCode == 'P')) {
        System.out.println("Invalid Package Code");
        System.out.print("Please enter package code (G/P): ");
        packageCode = Character.toUpperCase(keyboard.next().charAt(0));
      }

      System.out.print("\nPlease enter quantity of adults: ");
```

```
      adultQuantity = keyboard.nextInt();

      System.out.print("\nPlease enter quantity of children: ");
      childQuantity = keyboard.nextInt();

      // Calculation for ticket price
      if (packageCode == 'G') {
        ticketPrice = (adultQuantity * ADULT_GOLD_PRICE) + (childQuantity *
CHILD_GOLD_PRICE);
      } else {
        ticketPrice = (adultQuantity * ADULT_PLATINUM_PRICE) + (childQuantity
* CHILD_PLATINUM_PRICE);
      }

      // Confirmation for meal combo
      System.out.print("\nDo you want add on meal combo? (y/n): ");
      addOn = keyboard.next().equalsIgnoreCase("Y");

      // Get quantity and calculate charge
      if (addOn) {
        System.out.print("\nEnter quantity of meal combo: ");
        mealComboQuantity = keyboard.nextInt();

        // Calculate full meal combo charge
        mealComboCharge = mealComboQuantity * MEAL_COMBO_PRICE;
      }

      // Calculate final @ total charge
      total = ticketPrice + mealComboCharge;

      // Order Summary
      System.out.println("\nETS TICKET INFORMATION - Butterworth to Kuala
Lumpur");
      System.out.printf("PASSENGER ID:\t\t%s\n", passengerID);
      System.out.printf("Number of Adults:\t%d\n", adultQuantity);
      System.out.printf("Number of Childrens:\t%d\n", childQuantity);
      System.out.printf("Ticket Price:\t\tRM%,.2f\n", ticketPrice);
      System.out.printf("Meal Combo Price:\tRM%,.2f\n", mealComboCharge);
      System.out.printf("Net Ticket Price:\tRM%,.2f\n", total);

      System.out.print("\nisRepeat for a different Customer? (y/n): ");
      isRepeat = Character.toUpperCase(keyboard.next().charAt(0)) == 'Y';
      keyboard.nextLine();
    } while (isRepeat);
    keyboard.close();
  }
}
```

Question 2

```
import java.util.Scanner;

public class Canopy {
  public static double studentCharge = 90.00;
  public static double staffCharge = 150.00;
  public static double outsiderCharge = 200.00;

  public static void main(String[] args) {
    boolean isContinue;
    Scanner keyboard = new Scanner(System.in);
    do {
      char participantCode = 'N';
      int totalCanopies = 0;
      int totalDays = 0;
      boolean isSellingWetItem = false;
      double totalCharge = 0;

      // Table
      System.out.printf("| Participant Type | Participant Code | Charges / Day
|");
      System.out.printf("\n| ---------------- | ---------------- | -----------
-- |");
      System.out.printf("\n| Students         | T                | RM%.2f
|", studentCharge);
      System.out.printf("\n| Staff            | S                | RM%.2f
|", staffCharge);
      System.out.printf("\n| Outsider         | O                | RM%.2f
|", outsiderCharge);

      // Get Code
      System.out.print("\nPlease enter your code: ");
      participantCode = Character.toUpperCase(keyboard.next().charAt(0));

      // Loop for invalid code
      while (participantCode != 'T' && participantCode != 'S' &&
participantCode != 'O') {
        System.out.print("\nPlease enter your code: ");
        participantCode = Character.toUpperCase(keyboard.next().charAt(0));
      }
      keyboard.nextLine(); // Consume leftover \n

      // Get number of canopies
      System.out.print("\nPlease the number of canopies: ");
      totalCanopies = keyboard.nextInt();
      keyboard.nextLine(); // Consume leftover \n

      // Get day count
      System.out.print("\nPlease the number of days to use: ");
      totalDays = keyboard.nextInt();
      keyboard.nextLine(); // Consume leftover \n

      // Get wet item
      System.out.print("\nWill you be selling wet items? (y/n): ");
      isSellingWetItem = (Character.toUpperCase(keyboard.next().charAt(0)) ==
'Y');
```

```
      // Calculate total charge
      totalCharge = canopyCharge(participantCode, totalCanopies, totalDays)
          + additionalCharge(isSellingWetItem);

      // Summary
      System.out.println("\nParticipant Code:\t" + participantCode);
      System.out.println("Number of Canopies:\t" + totalCanopies);
      System.out.println("Selling Wet Item:\t" + isSellingWetItem);
      if (isSellingWetItem)
        System.out.println("Wet Item Charge:\t" +
additionalCharge(isSellingWetItem));
      System.out.println("Total Charge:\t\t" + totalCharge);

      // Continue?
      System.out.print("\nDo you want to continue for another customer? (y/n):
");
      isContinue = (Character.toUpperCase(keyboard.next().charAt(0)) == 'Y');
    } while (isContinue);
    keyboard.close();
  }

  public static double additionalCharge(boolean isSellingWetItem) {
    return (isSellingWetItem) ? 100 : 0;
  }

  public static double canopyCharge(char participantCode, int totalCanopies,
int totalDays) {
    switch (participantCode) {
      case 'T':
        return (studentCharge * totalCanopies) * totalDays;
      case 'S':
        return (staffCharge * totalCanopies) * totalDays;
      default:
        return (outsiderCharge * totalCanopies) * totalDays;
    }
  }
}
```